

Application of Information & Communication Technologies

Lecture-20

Recap of Lecture 19

- ◆ Introduction to Programming Languages
- ◆ Learning Objective:
 - Understand what programming languages are, their types, and how different languages express the same logic.

Overview of Lecture 20

- ◆ Computer communication Concepts:
 - Meaning and Importance of Computer Communication
 - Components of Communication System
 - Sending Device
 - Channel
 - Receiving Device

Computer Communication

- Computer communication means sharing data and information between two or more computers.
- It enables users to exchange messages, files, or commands electronically.
- 💡 Example: Sending an email, video calling, or browsing a website.



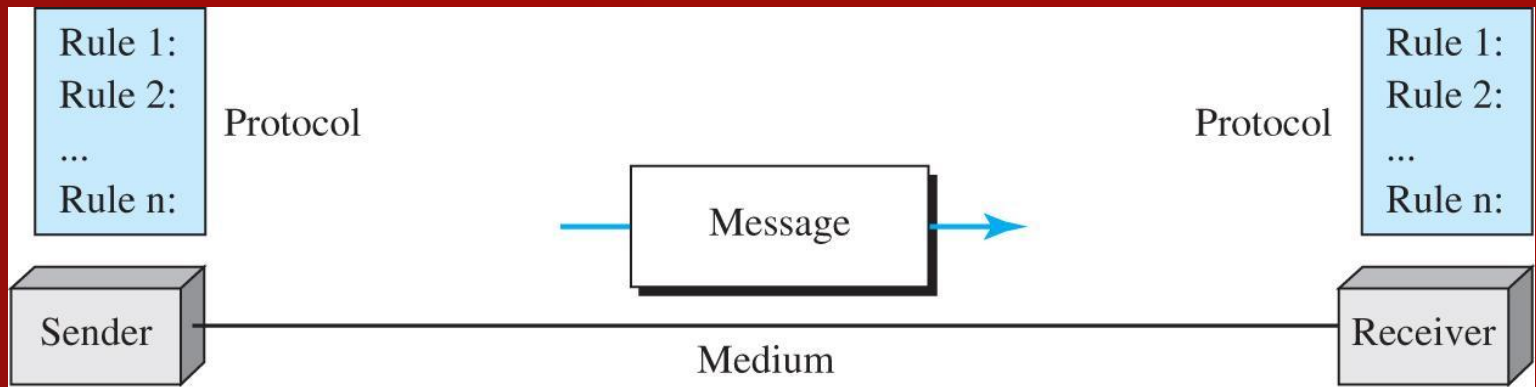
Figure 8-1 An example of a communications system. The communications channel consists of telephone and power lines, cable television and other underground lines, microwave stations, and satellites.

Importance of Computer Communication

- ◆ Enables fast and reliable **data exchange**.
- ◆ Reduces cost of physical meetings and travel.
- ◆ Improves **collaboration** and productivity.
- ◆ Allows **remote access** and cloud services.
- ◆ Forms the base for **internet and global networks**.

Components of a Communication System

- ◆ A data communications system has five components:



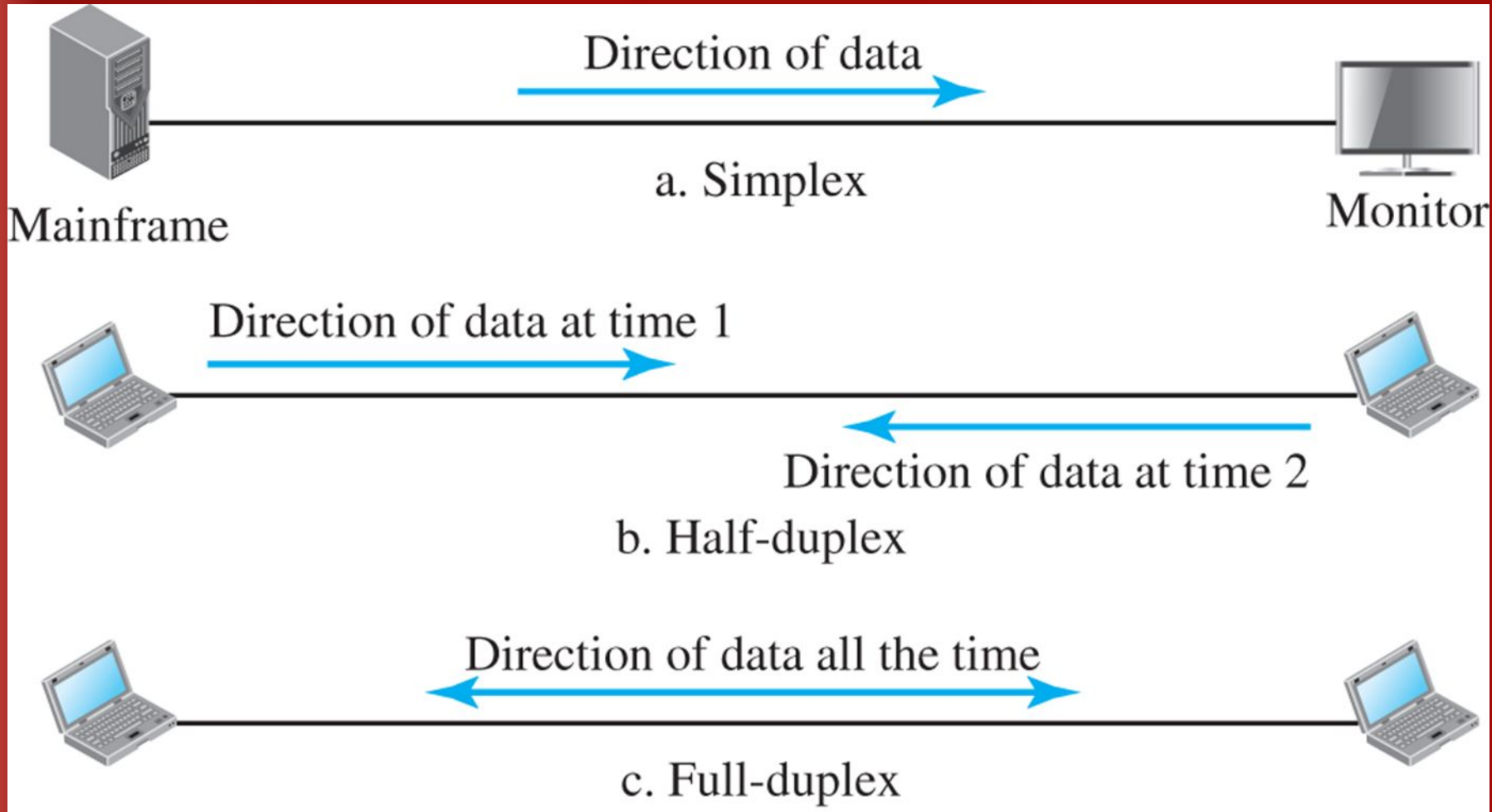
Components of a Communication System

- **A data communications system has five components:**
 - **Message:**
 - The actual information or data to be transmitted.
 - Example: text, audio, video, or file.
 - **Sender**
 - The device that creates and sends the message.
 - Example: computer, mobile phone, or sensor.
 - **Receiver**
 - The device that receives and interprets the message.
 - Example: server, printer, or another computer.
 - **Transmission Medium**
 - The path through which data travels.
 - Example: wired (fiber optic, cable) or wireless (radio, Wi-Fi).
 - **Protocol**
 - A set of rules that govern data communication.
 - Defines how data is formatted, transmitted, and received.
 - Example: TCP/IP, HTTP, FTP.

Communication Modes

- ◆ Also known as Data Flow
- ◆ Communication between two devices can be simplex, half-duplex, and duplex

Communication Modes

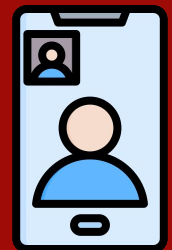
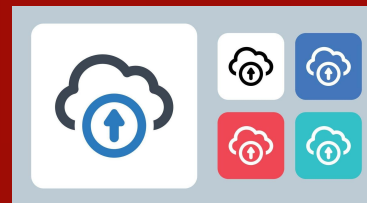
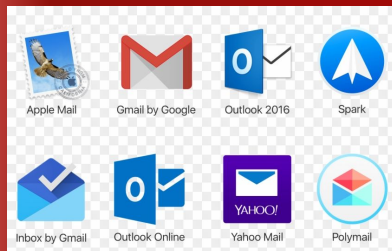


Comparison of Data Transmission Modes

Feature	Simplex	Half-Duplex	Full-Duplex
Direction of Data Flow	One-way only	Two-way, but one direction at a time	Two-way, simultaneously
Communication Example	Keyboard → Monitor	Walkie-Talkie	Telephone Call
Transmission Control	Only sender transmits; receiver only receives	Each device can both send and receive, but not at the same time	Both devices send and receive at the same time
Speed / Efficiency	Slow (limited)	Moderate	Fast and efficient
Cost / Complexity	Simple and inexpensive	Moderate	More complex and expensive
Use Case Examples	Display units, sensors	CB radio, data link between two stations	Video conferencing, modern LANs

Examples of Communication Systems

- ◆ Email communication
- ◆ Online chat/messaging
- ◆ Cloud storage sharing
- ◆ IoT devices (smart home sensors)
- ◆ Video conferencing platforms



Summary

- Communication allows computers to share data efficiently.
- Requires sender, medium, and receiver.
- Can be simplex, half duplex, or full duplex.
- Foundation for networking and the Internet.

Suggested Reading

- ◆ Ch-07, “Understanding Computers: Today and Tomorrow, Comprehensive”, 15th Edition by Deborah Morley & Charles S. Parker
- ◆ Ch-08, Discovering Computers Fundamentals-Your Interactive Guide -- Gary B Shelly; Misty E Vermaat; Jeffrey J Q